Lab 5.2: Breaking WEP and WPA and Decrypting the Traffic

Lab 5.2: Part 1 – Viewing Wireless Networks and Connected Devices

1. Research and describe with the command **aircrack-ng** is for

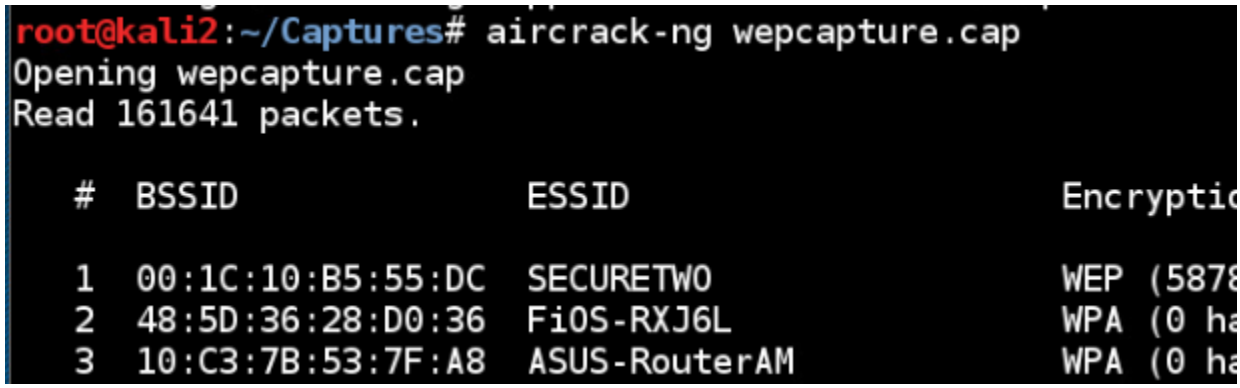
   The aircrack-ng command is used to crack WEP keys and WPA passphrases

2. Research and describe with the command **aircdecap-ng** is for

   The aircdecap-ng command is used to decrypt the WEP and WPA traffic

Lab 5.2: Part 2 – Cracking WEP

1. Question 10

2. 
```
root@kali2:~/Captures# aircrack-ng wepcapture.cap
Opening wepcapture.cap
Read 161641 packets.

   #  BSSID              ESSID                     Encryptic

   1  00:1C:10:B5:55:DC  SECURETWO                 WEP (5878
   2  48:5D:36:28:D0:36  FiOS-RXJ6L                WPA (0 ha
   3  10:C3:7B:53:7F:A8  ASUS-RouterAM             WPA (0 ha
```

3. Question 11

4.
```
Index number of target network ? 1

Opening wepcapture.cap
Attack will be restarted every 5000 captured ivs.
Starting PTW attack with 58781 ivs.
```

5. Question 12

6.
```
  KEY FOUND! [ 39:B0:35:D5:9C ]
```

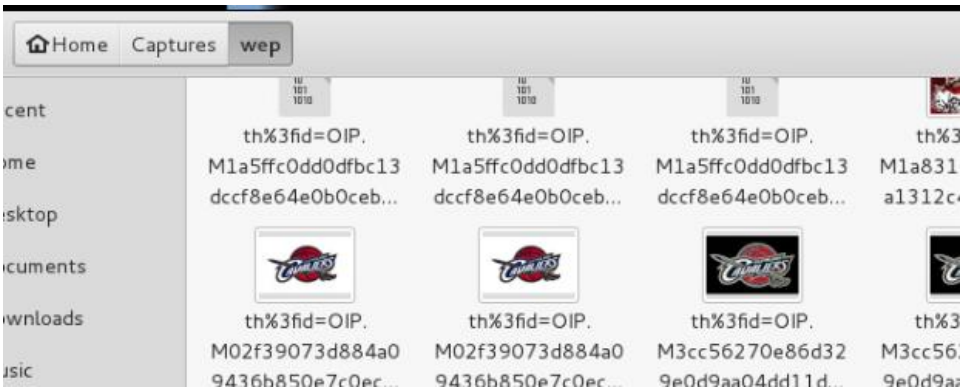
7. Question 13

8.
```
root@kali2:~/Captures# airdecap-ng -w 39:B0:35:D5:9C wepcapture.cap
Total number of packets read        161641
Total number of WEP data packets     74071
Total number of WPA data packets      1663
Number of plaintext data packets         0
Number of decrypted WEP  packets     74071
```

9. Question 24

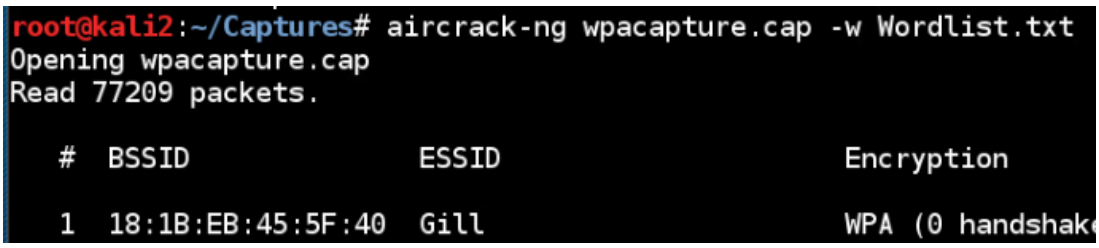
10.
```
------=_NextPart_000_0031_01D1A939.C9656330
Content-Type: text/plain;
.charset="us-ascii"
Content-Transfer-Encoding: 7bit

The Cleveland Cavs will win the NBA title.
-Reggie

------=_NextPart_000_0031_01D1A939.C9656330
```

11. Question 38

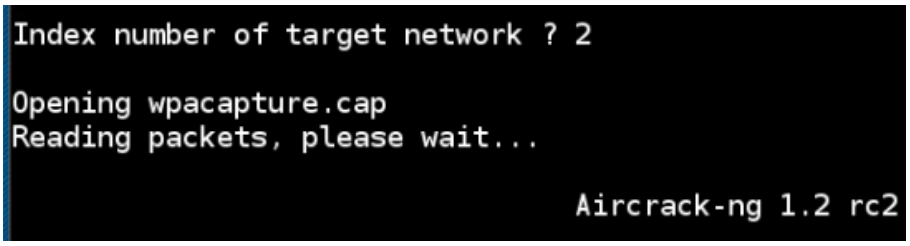


Lab 5.2: Part 3 – Cracking WPA

1. Question 6

2. 

3. Question 7

4. 

5. Question 8

6.

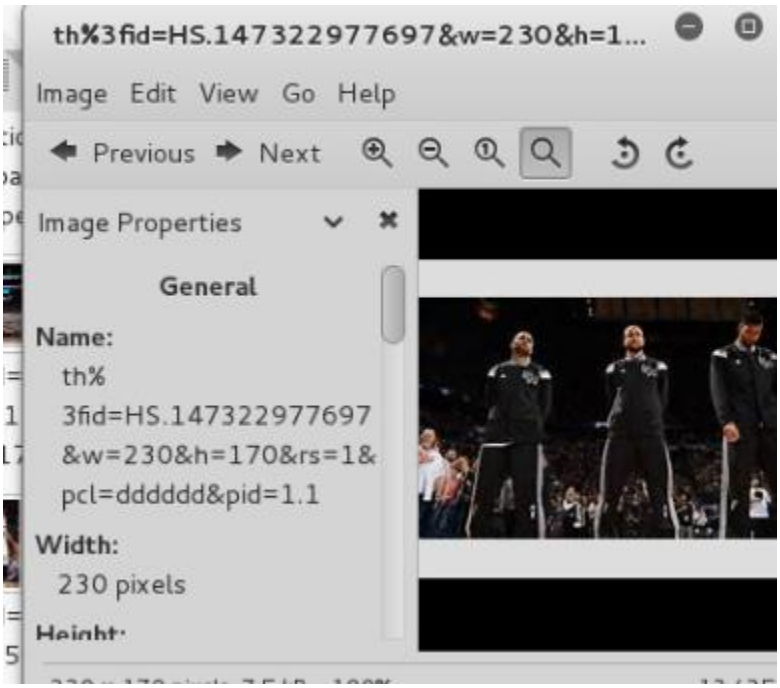
7. Question 9

8.
```
root@kali2:~/Captures# airdecap-ng -e SECURETWO -p boneless wpacapture.cap
Total number of packets read         77209
Total number of WEP data packets         0
Total number of WPA data packets     27913
Number of plaintext data packets        11
```

9. Question 17

10.
```
From: Administrator [mailto:Administrator@xyzcompar
Sent: Sunday, May 08, 2016 3:13 PM
To: administrator@xyzcompany.com
Subject: NBA


I think the Spurs will win the NBA title

------= NextPart_000_003F_01D1A943.883458D0
```

11. Question 25



Lab 5.2: Part 4 – Lessons Learned

Lessons Learned

Learned

- In this lab, I was walked through the process of how to crack WEP keys and WPA passphrases and then decrypt the traffic that went through WEP and WPA. I got more familiar with Wireshark by using the filters again and finding decrypted information in the TCP stream. After completing the WEP part of the assignment, the WPA was easy, and I felt I was getting the hang of the process. Many of the actions done to the WEP portion of the assignment were the same or similar to the actions needed to do the WPA. I also learned that the encryption used for WEP is no longer used as it is weak and can be cracked easily. Now, we use WPA2 and WPA3 which use AES encryption which is much more secure than the RC4 encryption used in WEP and WPA.

Surprises & Challenges

- This lab was easy to follow and understand through each of the steps and I did not have any challenges along the way.
- I do wish we were able to capture the network traffic and save it to our system then be able to crack the passphrases and decrypt the data so we could get the feel of the whole process.

Feedback

- This lab took me about an hour to complete
- My favorite part of the lab was watching the WPA passphrase be found as it ran through multiple passwords until it ended up at the correct one